Lakshya JEE 2.0 2025

Mathematics

DPP: 1

Indefinite Integration

Q1 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{1-\tan^2 x}{1+\tan^2 x} dx$

Q2 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{e^{5 \ln x}-e^{4 \ln x}}{e^{3 \ln x}-e^{2 \ln x}} dx$$

Q3 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int\left(e^{a \ln x}+e^{x \ln a}\right) dx(a>0)$

Q4 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{(1+x)^2}{x(1+x^2)} dx$

Q5 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{e^{2x}-1}{e^x} dx$

Q6 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{\sin x+\cos x}{\sqrt{1+\sin 2x}} dx(\cos x+\sin x>0)$$

Q7 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{\cos 2x-\cos 2\alpha}{\cos x-\cos \alpha} dx$

Q8 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \sqrt{1-\sin 2x} dx$

Q9 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$

Q10 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{1+2x^2}{x^2(1+x^2)} dx$

Q11 $\int \frac{x^4+x^2+1}{2(1+x^2)} dx=f(x)+c$, where c is an integration constant then $f(1)$ is equal to:
(A) $\frac{1}{6}$
(B) $\frac{\pi}{8}+\frac{1}{6}$
(C) $\frac{\pi}{8}+\frac{1}{2}$
(D) $\frac{\pi}{4}+\frac{1}{6}$

Q12 $\int 2^x \cdot e^x dx=$
(A) $\frac{(2e)^x}{\ln(2e)}+c$
(B) $\frac{e^2}{\ln(2e)}+c$
(C) $\frac{(2e)^x}{\ln 2x}+c$
(D) None of these

Q13 $\int \frac{\sqrt{x^4+x^{-4}+2}}{x^3} dx=$
(A) $\ln|x|+\frac{x^{-4}}{4}+c$
(B) $\ln|x|-\frac{x^{-4}}{4}+c$
(C) $\ln x-\frac{x^3}{3}+c$
(D) None of these

Q14 $\int \frac{e^{3x}+e^{5x}}{e^x+e^{-x}} dx=g(x)+k$, where k is an integration constant then $g(1/4)$ equals:

(A) $\frac{e^4}{4}$
(B) e^4
(C) $\frac{e}{4}$
(D) e

Q15 $I=\int \frac{1+\tan^2 x}{1+cot^2 x}\, d\, x=A\, x+B\ \tan\ x+C,$
$where\ A,\ B\, and\ \ C\, are\, cons\tan ts,\ then$
$A+B\, equals\, to:$

(A) 0
(B) 1
(C) 2
(D) 3

Q16 $\int \frac{2^{x+1}-5^{x-1}}{10^x} dx=\frac{2\cdot a^x}{\ln a}-\frac{1}{5}\frac{b^x}{\ln b}+c$, where c is an integration constant then $10ab=$

(A) -1
(B) 0
(C) 1
(D) 2

Q17 $\int \left(x^2+5\right)^3 dx$

Q18 $\int \frac{x^4+x^2+1}{x^2-x+1} dx$ is equal to
(A) $\frac{x^3}{3}-\frac{x^2}{2}+x+C$
(B) $\frac{x^3}{3}+\frac{x^2}{2}+x+C$
(C) $\frac{x^3}{3}-\frac{x^2}{2}-x+C$
(D) $\frac{x^3}{3}+\frac{x^2}{2}-x+C$

Q19 If $g(x)$
$=\left(4\cos^4 x-2\cos 2x-\frac{1}{2}\cos 4x-x^7\right)^{1/7}$
then $\int g(g(x))dx$ is:
(A) $2x+c$
(B) $\frac{x^2}{2}+c$
(C) $-2\cos x+c$
(D) $2\sin x+c$

Q20 Which one of the following is true?
(A) $x\cdot\int \frac{dx}{x}=x\ln|x|+C$
(B) $x\cdot\int \frac{dx}{x}=x\ln|x|+Cx$
(C) $\frac{1}{\cos x}\cdot\int \cos x dx=\tan x+C$
(D) $\frac{1}{\cos x}\cdot\int \cos x dx=x+C$

Q21 $\int \frac{x^2+5x-1}{\sqrt{x}} dx$

Answer Key

Q1 $I=\frac{\sin 2x}{2}+c$

Q2 $\frac{x^3}{3}+C$

Q3 $I=\frac{x^{a+1}}{a+1}+\frac{a^x}{\ln a}+c$

Q4 $\ln x+2\tan^{-1}x+C$

Q5 $e^x+e^{-x}+C$

Q6 x + C

Q7 $2(\sin x+x\cos\alpha)+C$

Q8 $(\sin x+\cos x)\, sgn\,(\cos x-\sin x)+C$

Q9 $-\cot x-\tan x+c$

Q10 $I=\tan^{-1}x-\frac{1}{x}+c$

Q11 (B)

Q12 (A)

Q13 (B)

Q14 (C)

Q15 (A)

Q16 (C)

Q17 $=\frac{x^7}{7}+3x^5+25x^3+125x+C$

Q18 (B)

Q19 (B)

Q20 (B)

Q21 $\frac{2}{5}x^{5/2}+\frac{10}{3}x^{3/2}-2x^{1/2}+c$